



CHALLENGE 04 / 20

THE DUPLICATE ADMINISTRATOR

IDENTITY LINEAGE AND ACCOUNT-ORIGIN FORENSICS

NATIONAL LEVEL HACKATHON • 2026

POWERED BY SYNTHOQUEST PVT LTD

Student Project Development Challenge • No organizer-provided evidence package.

1 | PROBLEM BACKGROUND

A national logistics company with 60,000 staff, three acquired subsidiaries, four directory forests, a cloud identity provider, a privileged access vault and 140 SaaS applications found, during an audit, a second Domain Administrator with the same name, alias and privileges as an existing one. It has authenticated almost daily for eleven months. No ticket or approval exists for it.

Two mergers re-keyed identities, the HR system was migrated twice, and a federation trust with a former subsidiary still exists. Three identity engineers have since left. Governance records were overwritten during migration. The account could be a synchronisation artefact, a legitimate delegate, a clone, a synthetic identity or an attacker implant, and it may have changed hands during its life.

Disabling it may break a production workflow. Leaving it may leave an attacker in place.

2 | PROBLEM STATEMENT

Develop an **Identity Lineage Reconstruction System** that establishes how an account came to exist, who controlled it at every point in its life and whether control changed hands, across directories, federation, privileged access and SaaS systems that each hold a partial, inconsistent view.

The system must classify the account's origin and predict the operational impact of disabling it, without testing in production.

Hard constraints

- Identity systems disagree on identifiers, and merges have rewritten history.
- An account may have been legitimately created and later taken over; the handover must be found.
- A malicious clone that mimics synchronisation behaviour must be separated from a real synchronisation artefact.
- The system must work across tens of thousands of identities and several forests.
- Impact of disabling the account must be estimated from evidence of what depends on it.

No evidence package or dataset is provided. Teams decide what to build, what data they need, how to simulate the environment and how to validate the result. No tools or architecture are prescribed.

3 | MANDATORY FEATURES

Teams must develop a working software solution. The system must provide the following features. How each is achieved is for the team to decide.

01	Cross-System Identity Resolution	Ability to link one identity across forests, federation and SaaS.
02	Creation Pathway Recovery	Ability to establish how and by whom an account was created.
03	Control Handover Detection	Ability to detect a change in who controls an account.
04	Sync-Artefact Discrimination	Ability to separate synchronisation artefacts from malicious clones.
05	Privilege Lineage Tracing	Ability to trace how privileges were granted, inherited or inflated.
06	History-Rewrite Handling	Ability to reason when merges and migrations rewrote history.
07	Dependency Impact Estimation	Ability to estimate what breaks if the account is disabled.
08	Lifecycle Explanation	Ability to present the full account lifecycle for review.

The system must clearly distinguish between:

DOCUMENTED	DELEGATED	CLONED	SYNTHETIC	UNAUTHORISED
------------	-----------	--------	-----------	--------------

An account origin that cannot be supported must not be presented as established.

Build, don't search. This is a development challenge. Teams must create the system and the controlled environment that proves it works. Submitting an analysis of a given dataset is not a solution.

4 | EXPECTED OUTPUT

Live demonstration. The team must show that its system can handle a complex controlled scenario created by the team. The demonstration must show:

01	Teams' own simulated multi-forest, federated environment
02	Several kinds of duplicate account planted in it
03	Recovery of creation pathway
04	Detection of a control handover
05	Separation of a clone from a sync artefact
06	Privilege lineage
07	Estimated impact of disabling the account

The team must be able to explain why the system reached each major conclusion.

Validation standard (every team)

- Build a scenario generator with known ground truth and evaluate on at least ten distinct scenarios, including ones with deliberately misleading evidence.
- At the demonstration, judges change scenario parameters on the spot. The system must adapt without being rebuilt.
- Include an adversary variant that actively tries to defeat the system, and report how the system held up.
- Stated confidence must match measured accuracy. Overconfident or unexplained conclusions lose marks.
- No real systems, real personal data or functional malware may be used.

Submission. Each team must submit:

01	Working Prototype	A functional implementation of the proposed solution.
02	Source Code	Complete source code required to reproduce the project.
03	Technical Architecture	Architecture and component description.
04	Validation Environment	A controlled environment created by the team to demonstrate the solution.
05	Technical Documentation	Implementation, design and testing documentation.
06	Forensic Investigation Report	A professional report generated from the demonstrated investigation.
07	Demonstration	Live technical demonstration to the judging panel.
08	Limitations	Known limitations and situations where the system cannot establish a reliable conclusion.